

Functional Annotation Report: rbsR (PP_2457, UniProt Q88K35)

Pseudomonas putida KT2440 (strain ATCC 47054 / DSM 6125 / NCIMB 11950 / KT2440)

0. Gene identity verification (mandatory)

The target was cross-checked against UniProt Q88K35 before any conclusions were drawn:

- **Gene symbol / locus:** `rbsR` / `PP_2457` — consistent with UniProt and KEGG (`ppu:PP_2457`, GenBank AAN68069).
- **Protein description:** "DNA-binding transcriptional repressor" — matches the LacI-family domain content found below.
- **Organism:** *Pseudomonas putida* KT2440 (NCBI TaxID 160488) — correct.
- **Domains match:** UniProt/InterPro list HTH_LacI (IPR000843), λ-repressor-like DNA-binding sf (IPR010982), and the periplasmic-binding-protein / LacI sugar-binding domain (IPR001761, PF00356 LacI). These are exactly the domains observed in the sequence analysis. **Identity confirmed.**

The name `rbsR` is used across many bacteria for the *ribose operon repressor*. Here it is **not** a spurious name transfer: the *P. putida* gene sits inside a genuine ribose-utilization operon (Section 3), so the annotation is internally consistent for this organism.

1. Summary (answer to the research question)

rbsR (PP_2457) encodes RbsR, a cytoplasmic LacI/GalR-family, one-component transcriptional repressor that controls the ribose (rbs) regulon of *P. putida* KT2440. Its primary function is to bind operator DNA in the promoter region of the adjacent ribose transport/catabolism genes and repress their transcription; the C-terminal sugar-binding domain senses the pathway substrate (D-ribose, or its phosphorylated derivative ribose-5-

phosphate), and effector binding relieves repression so that ribose can be imported and catabolized. It is a DNA-binding regulatory protein, not an enzyme or transporter, and it acts in the cytoplasm on the nucleoid.

2. Molecular identity and domain architecture

RbsR is a 340-residue protein (36.65 kDa; UniProt Q88K35) with the canonical **two-domain LacI/GalR architecture**:

Region	Domain	Evidence
~2–56 (N-terminus)	lacI-type helix-turn-helix DNA-binding domain	PROSITE PS50932; Pfam PF00356 (LacI); SMART SM00354; CDD cd01392 (HTH_LacI); PRINTS PR00036 (HTHLACI)
~60–340 (C-terminus)	Periplasmic-binding-protein-like effector/sugar-binding domain	Pfam PF00532 (Peripla_BP_1); InterPro IPR001761; SUPFAM SSF53822 (Periplasmic binding protein-like I)

Orthology assignments are mutually consistent: **COG1609** (LacI-family transcriptional regulator), **KEGG K02529**, and **PANTHER PTHR30146 / SF148** ("LacI-related transcriptional repressor," PurR-related). This family (LacI, GalR, PurR, RbsR) works by the same logic: the HTH reads an operator, and the sugar-binding domain binds a metabolite that allosterically switches DNA affinity [PMID 23651393].

The presence of both a DNA-binding domain and a sugar-binding domain marks RbsR as a **sugar-responsive one-component transcription factor** rather than a metabolic enzyme.

Quantitative homology. Global pairwise alignment (Needleman–Wunsch, BLOSUM62) of PP_2457 against characterized *E. coli* LacI-family repressors places it clearly in the **RbsR/PurR subclade**:

Reference (E. coli)	% identity to PP_2457
RbsR (P0ACQ0)	43.4% (142/327)
PurR (P0ACP7)	45.8% (154/336)
GalR (P03024)	35.8% (121/338)
LacI (P03023)	30.6% (103/337)

PP_2457 is ~8–15 points more similar to the RbsR/PurR pair than to LacI or GalR, matching its "PurR-related" PANTHER annotation. Because RbsR and PurR are each other's closest paralogs, **sequence alone cannot decide ribose vs. purine specificity** — that is resolved by genomic context (Section 3), which places PP_2457 inside the ribose operon, not a purine-biosynthesis locus.

3. Biological role: repressor of the ribose regulon (organism-specific evidence)

The strongest *P. putida*-specific evidence is **conserved gene neighborhood (synteny)**. rbsR (PP_2457, chromosomal coordinates 2,804,188–2,805,210) is embedded in a contiguous ribose-utilization gene cluster:

Locus	Gene	Product / function
PP_2454	rbsB	Ribose ABC transporter, periplasmic ribose-binding protein
PP_2455	rbsA-I	Ribose ABC transporter, ATP-binding subunit
PP_2456	rbsC	D-ribose ABC transporter, permease subunit
PP_2457	rbsR	DNA-binding transcriptional repressor (this gene)
PP_2458	rbsK	Ribokinase (EC 2.7.1.15): D-ribose → D-ribose-5-phosphate
PP_2459	rbsD	Ribose pyranase (anomerase, furanose/pyranose interconversion)
PP_2460	nuh	Ribonucleoside hydrolase (nucleoside → base + ribose)

This layout mirrors the canonical *E. coli* `rbsDACBK` operon in which `rbsR` is the adjacent LacI-type repressor [PMID 23651393]. Functionally, this cluster encodes a complete ribose-scavenging pathway: high-affinity ribose uptake (RbsABC), release of ribose from nucleosides (Nuh), and conversion of internalized D-ribose to ribose-5-phosphate (RbsD pyranase to open/interconvert anomers, RbsK ribokinase to phosphorylate) for entry into the pentose phosphate pathway. RbsR is the local regulator of this system.

4. Mechanism (transferred from characterized orthologs)

No dedicated biochemical study of the *P. putida* protein exists in the literature searched; the mechanism is inferred from well-characterized RbsR orthologs that share both the domain architecture and the operon context:

- **Repression by operator binding.** In *E. coli*, "in the absence of inducer d-ribose, the ribose operon is repressed by a LacI-type transcription factor RbsR" [PMID 23651393].
- **Operator is an inverted repeat; D-ribose is the effector.** In *Bifidobacterium breve* UCC2003, "the promoter upstream of `rbsABCDK` is negatively controlled by RbsR(His) binding to an 18 bp inverted repeat and... RbsR(His) binding activity is modulated by D-ribose" [PMID 21255330].
- **Effector may be the phosphorylated sugar.** In *Corynebacterium glutamicum*, "a probable negative effector of RbsR in vivo is ribose 5-phosphate or a derivative thereof" [PMID 19118356].

Predicted mechanism for PP_2457: Under ribose-poor conditions RbsR homodimers/tetramers occupy an operator (likely a palindromic inverted repeat) in the `rbs` promoter region and block transcription of the uptake and catabolic genes. When D-ribose (or ribose-5-phosphate) accumulates, the effector binds the C-terminal sugar-binding domain, driving the LacI-type conformational change that reduces operator affinity and **derepresses** the operon, licensing ribose consumption. The specific effector (free ribose vs. ribose-5-P) and the operator sequence in *P. putida* remain to be determined experimentally.

5. Subcellular localization

RbsR is a **cytoplasmic** protein. It has no signal peptide or transmembrane segments (UniProt lists only the N-terminal HTH domain feature), and its function—sequence-specific operator binding—takes place on chromosomal DNA in the cytoplasm/nucleoid. This contrasts with its operon partner RbsB, which is a periplasmic solute-binding protein; RbsR itself does not leave the cytosol.

6. Pathway placement

- **Pathway:** D-ribose catabolism / pentose utilization, feeding the **pentose phosphate pathway** via ribose-5-phosphate.
 - **Regulatory node:** transcriptional control (repression, ribose-relieved) of the rbs uptake+catabolism operon.
 - **Broader (secondary) note:** In *E. coli*, RbsR has been reported as a wider regulator linking ribose availability to purine-nucleotide de novo synthesis vs. salvage [PMID 23651393]. Whether the *P. putida* ortholog has a comparably expanded regulon is unknown and would be a pleiotropic extension beyond its core, synteny-supported role; it is flagged here only for completeness.
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7. Supported vs. refuted hypotheses

- **Supported (high confidence):** RbsR is a LacI-family DNA-binding transcriptional repressor (domain evidence, direct). ✓
 - **Supported (strong inference):** RbsR represses the adjacent ribose transport/catabolism operon (operon synteny + ortholog function). ✓
 - **Supported (inference):** Repression is relieved by a ribose-derived effector binding the C-terminal domain (ortholog mechanism). ✓
 - **Not supported / not applicable:** RbsR is an enzyme, transporter, or structural protein — refuted by domain architecture and family assignment. ✗
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8. Evidence quality and limitations

- **Direct evidence** exists for the *molecular class* (curated domain/family databases: Pfam, InterPro, PROSITE, COG, KEGG).
- **Organism-specific functional evidence** is **inferential**, resting on (a) conserved gene neighborhood in *P. putida* KT2440 and (b) experimentally characterized orthologs in *E. coli*, *B. breve*, and *C. glutamicum*. No *P. putida*-specific gene-knockout, EMSA, or expression study of PP_2457 was found in the literature searched.
- **Open questions:** the exact operator sequence, the physiological effector (D-ribose vs. ribose-5-phosphate), the oligomeric state, and whether the regulon extends beyond the rbs cluster in *P. putida*. Targeted experiments (rbsR deletion + RNA-seq, purified-protein EMSA/ DNase footprinting, effector titration) would convert these inferences into direct evidence. RB-TnSeq fitness data (LBL Fitness Browser) for *P. putida* likely bear on this but were not retrievable during this analysis (access restricted).

References

- [P 23651393](#)
— Shimada, Kori, Ishihama (2013). RbsR and the *E. coli* rbsDACBK operon; ribose-relieved repression; global role in purine metabolism.
- [P 21255330](#)
— Pokusaeva et al. (2010). *B. breve* RbsR represses rbsABCDK via an 18-bp inverted repeat, modulated by D-ribose.
- [P 19118356](#)
— Nentwich et al. (2009). *C. glutamicum* RbsR; effector is ribose-5-phosphate or a derivative.
- [P 16519689](#)
— Müller et al. (2006). *B. subtilis* RbsR as a LacI/GalR-family ribose operon repressor.
- UniProt Q88K35; KEGG ppu:PP_2457; InterPro/Pfam domain assignments (accessed 2026).

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